

kinetic-kit

API Reference

v0.1.1

This document lists all public symbols exported by kinetic-kit.

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1 Template

`thesis()` is the main entrypoint of the package. By default, it assembles a complete doctoral thesis in the KSP-approved style. The template is fully configurable and can also be used to produce a bachelor's or master's thesis. Note that the KSP endorsement only applies to doctoral theses in the default configuration; customized documents are not covered by it.

thesis

The official Typst template for doctoral theses published through KIT Scientific Publishing (KSP).

Parameters

```
thesis(  
  format: str,  
  lang: str,  
  author-firstname: str,  
  author-surname: str,  
  title: content,  
  title-page: content or function or none,  
  front-matter: content or none,  
  appendix: content or none,  
  back-matter: content or none,  
  serif-headings: bool,  
  heading-numbering-depth: int,  
  figure-kinds: array,  
  margin-preset: str,  
  binding-correction: length,  
  colored-links: bool,  
  draft: bool,  
  draft-info: str or none,
```

```
doc: content
) -> content
```

format `str`

Paper format — one of "a5" (148×210 mm, default), "17x24" (170×240 mm), or "a4" (210×297 mm, discouraged by KSP). Font sizes and margins are set automatically.

Default: "a5"

lang `str`

Document language — "de" or "en".

Default: "de"

author-firstname `str`

Author's first name.

Default: "Max"

author-surname `str`

Author's surname.

Default: "Mustermann"

title `content`

Thesis title.

Default: [Your Thesis Title]

title-page `content or function or none`

Title page. Defaults to the built-in doctoral page; pass `doctoral-title-page.with(...)` to configure it, your own content, a function called as `(title, author-firstname:, author-surname:, format:, lang:)`, or none to omit it.

Default: `doctoral-title-page`

front-matter `content or none`

Roman-numeral pages before the body — abstracts, acknowledgements, the table of contents, any page of your own. Defaults to just `outlines.table-of-contents()`. Pass none for no front matter.

Default: `table-of-contents()`

appendix `content or none`

Appendix chapters. Template applies A, A.1, ... numbering and places the appendix directly after the body, before the back matter. none = omit.

Default: none

back-matter content or none

Pages after the appendix — list pages, bibliography, own publications, whatever the document needs, in the order written. none = omit. Refer to outlines for list-page helpers.

Default: none

serif-headings bool

Use serif font for headings when true. Default false (sans-serif).

Default: false

heading-numbering-depth int

Deepest heading level to number. Default 3.

Default: 3

figure-kinds array

Figure kinds beyond image, table and raw, as dictionaries with kind and supplement. supplement takes either one value or one per language, e.g. (de: [Algorithmus], en: [Algorithm]).

Declaring a built-in kind is an error.

Default: ()

margin-preset str

Margin profile keyed on page count — "short", "medium", or "long".

Default: "short"

binding-correction length

BCOR added to inside margin. Default 0mm.

Default: 0mm

colored-links bool

KIT Blue links when true, black when false.

Default: true

draft bool

Show "ENTWURF" watermark when true.

Default: false

draft-info str or none

Optional version string below watermark. Default none.

Default: none

doc content

Main document body (chapters only).

2 Style Constants

The kit-style dictionary exposes the template's visual constants so that custom figures, diagrams, and other content can match the document's typography and KIT color palette exactly.

```
#import "@preview/kinetic-kit:0.1.1": kit-style

#let font-sizes = kit-style.font-sizes-by-format.at("a5")
#set text(font: kit-style.fonts.sans, size: font-sizes.small)
#rect(fill: kit-style.colors.green15, stroke: kit-style.colors.green)
```

2.1 Fonts

Field	Value	Description
fonts.serif	("Libertinus Serif",)	Serif body text
fonts.sans	("Libertinus Sans",)	Sans-serif headings and headers
fonts.mono	("Libertinus Mono",)	Code listings
leading	0.75em	Paragraph line spacing ($\approx 1.15\times$ at 10 pt)

2.2 Font Sizes

font-sizes-by-format is a dictionary keyed by format string ("a5", "17x24", "a4"). Each entry contains the sizes below. Access them with `kit-style.font-sizes-by-format.at(format)`.

All fields are length. Example values shown for the "a5" format (10 pt base).

Field	A5 / 17x24	A4	Description
base	10pt	11pt	Body text
chapter	18pt	25pt	Chapter heading (level 1)
section	14pt	17pt	Section heading (level 2)
subsection	12pt	14pt	Subsection heading (level 3)
subsubsection	10pt	11pt	Subsubsection heading (level 4+)
small	8pt	9pt	Running headers, footnotes, captions
footnote	8pt	9pt	Footnote text
title	18pt	25pt	Title page — document title
author	14pt	14pt	Title page — author name
title-info	14pt	14pt	Title page — info lines

2.3 Colors

All values are color. KIT brand colors follow a systematic naming scheme: base name plus optional opacity suffix (70, 50, 30, 15 = 70 %, 50 %, 30 %, 15 % tint).

Field	Hex	Description
colors.green	#009682	KIT Green — primary brand color
colors.green70	#4CB5A7	
colors.green50	#7FCAC0	
colors.green30	#B2DFD9	
colors.green15	#D9EFEC	
colors.blue	#4664AA	KIT Blue — links and accents

Field	Hex	Description
colors.blue70	#7D92C3	
colors.blue50	#A2B1D4	
colors.blue30	#C7D0E5	
colors.blue15	#E3E8F2	
colors.black	#000000	
colors.black70	#4D4D4D	
colors.black50	#808080	
colors.black30	#B3B3B3	
colors.black15	#D9D9D9	
colors.palegreen	#82BE3C	KIT extended palette
colors.yellow	#FAE614	
colors.orange	#DCA01E	
colors.brown	#A08232	
colors.red	#A01E28	
colors.lilac	#A00078	
colors.cyanblue	#50AAE6	
colors.keyword	#0000C8	Syntax highlighting — keyword
colors.comment	#3F7F5F	Syntax highlighting — comment
colors.string	#700055	Syntax highlighting — string literal

3 Title Page

doctoral-title-page is the default title-page. It owns every parameter printed on the page. Pass `doctoral-title-page.with(...)` to configure it, or your own content / function for anything else.

doctoral-title-page

Render the KIT doctoral thesis title page (German legal format). All text on this page is always in German regardless of document language.

Parameters

```
doctoral-title-page(
  title: content,
  author-title: content,
  author-firstname: str,
  author-surname: str,
  author-male: bool,
  doc-degree: str,
  doc-degree-f: str,
  department: str,
  university-genitive: str,
  status-approved: bool,
  exam-date: content,
  main-advisor: content,
  main-advisor-male: bool,
```

```
co-advisor: content,  
co-advisor-male: bool,  
format: str,  
lang: str  
) -> content
```

title content

Dissertation title.

author-title content

Academic title preceding the author's name (none = omit).

Default: "M.Sc."

author-firstname str

Author's first name.

Default: "Max"

author-surname str

Author's surname.

Default: "Mustermann"

author-male bool

Selects grammatical gender for the degree article.

Default: true

doc-degree str

Degree name in masculine form (e.g. "Doktor-Ingenieur").

Default: "Doktors der Ingenieurwissenschaften (Dr.-Ing.)"

doc-degree-f str

Degree name in feminine form (e.g. "Doktor-Ingenieurin").

Default: "Doktorin der Ingenieurwissenschaften (Dr.-Ing.)"

department str

KIT department or faculty.

Default: "KIT-Fakultät für Maschinenbau"

university-genitive str

University name in genitive (e.g. "des Karlsruher Instituts...").

Default: "des Karlsruher Instituts für Technologie (KIT)"

status-approved `bool`

false = submitted version, true = approved version.

Default: `false`

exam-date `content`

Date of oral examination — shown only when status-approved is true.

Default: `none`

main-advisor `content`

Main referee — shown only when status-approved is true.

Default: `none`

main-advisor-male `bool`

Selects gendered label for the main advisor.

Default: `true`

co-advisor `content`

Co-referee — shown only when status-approved is true.

Default: `none`

co-advisor-male `bool`

Selects gendered label for the co-advisor.

Default: `true`

format `str`

Paper format — "a5", "17x24", or "a4". Determines font sizes and title-page margins.

Default: `"a5"`

lang `str`

Accepted for the title-page contract and ignored — this page is always German.

title, author-firstname, author-surname, format and lang are supplied by the template when this is passed to title-page; setting them via `.with()` has no effect there.

Default: `"de"`

4 Outlines

The outlines namespace holds the sections whose body the template generates: the table of contents and the back-matter list pages. Place them inside the front-matter / back-matter content of `thesis()`.

```
#import "@preview/kinetic-kit:0.1.1": outlines, thesis
```

```
#show: thesis.with(
```

```

front-matter: [
  = Kurzfassung
  ...
  #outlines.table-of-contents()
],
back-matter: [
  #outlines.list-of-figures()
  #outlines.list-of("algorithm", [Algorithmenverzeichnis])
  #bibliography("refs.bib", title: [Literaturverzeichnis], style: "ieee")
],
)

```

table-of-contents

Print the table of contents.

Parameters

```
table-of-contents(title: content or str or auto, depth: int) -> content
```

title content or str or auto

Heading of the outline. auto uses Typst's localized default for the document language.

Default: **auto**

depth int

Deepest heading level to list.

Default: **3**

list-of

Print a back-matter list page for one figure kind.

Handles the in-outline state that switches flex-caption to its short form, which a hand-rolled outline(target: ...) would miss.

Parameters

```
list-of(kind: function or str, title: content or str) -> content
```

kind function or str

Figure kind to list — an element function such as image, or a string such as "algorithm".

title content or str

Heading of the list page.

list-of-figures

Print the list of figures.

Parameters

```
list-of-figures(title: content or str or auto) -> content
```

title content or str or auto

Heading of the list page. auto uses the localized default for the document language.

Default: **auto**

list-of-tables

Print the list of tables.

Parameters

```
list-of-tables(title: content or str or auto) -> content
```

title content or str or auto

Heading of the list page. auto uses the localized default for the document language.

Default: **auto**

list-of-listings

Print the list of listings.

Parameters

```
list-of-listings(title: content or str or auto) -> content
```

title content or str or auto

Heading of the list page. auto uses the localized default for the document language.

Default: **auto**

5 Figure Captions

flex-caption

Two-part caption: short version for LoF/LoT, long version under the figure.

Usage: `#figure(..., caption: flex-caption(short: [Short], long: [Long.] )`

Parameters

```
flex-caption(short: content, long: content) -> content
```

short content

Short caption shown in List of Figures / Tables.

Default: **none**

long content

Full caption shown below the figure in the document body.

Default: **none**

6 Figure Kinds

Typst keeps a separate counter and supplement for every figure kind. The template carries strings for Typst's own image, table and raw. Any other kind — pseudocode, theorems, whatever a document needs — is declared through the `figure-kinds` parameter of `thesis()`, which supplies the caption supplement. Its list page, if you want one, is a `#outlines.list-of("algorithm", [List of Algorithms])` call in back-matter.

Each entry is a dictionary with `kind` and `supplement`. `supplement` takes either one value or one per language:

```
#show: thesis.with(  
  figure-kinds: (  
    (kind: "algorithm", supplement: (de: [Algorithmus], en: [Algorithm])),  
    (kind: "theorem",    supplement: [Theorem]),  
  ),  
)
```

Per-chapter counter resets are automatic: they are derived from the figures present in the document, so every kind restarts each chapter whether or not it has been declared.